

Introduction and Partial Methods

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Introduction

In California, the health of patients is threatened by emergency department (ED) overcrowding. To address ED overcrowding, a hospital can close and redirect ambulances to other hospitals in a process called ambulance diversion by implementing diversion status. Data on hospital diversion status from ED-containing hospitals in California were gathered by the California Office of Statewide Health Planning and Development. In particular, the dataset spans three years, from 2013 to 2015, and includes 10 variables: **year** (the year; 2013, 2014, or 2015), **id** (a numerical label for each hospital), **county** (the California county each hospital belongs to), **trauma** (each hospital's trauma center designation; None/0, Level I, Level II, or Level III), **totalvisits** (how many EMS visits were made to the hospital in that year), **admit** (how many admitted EMS visits occurred for the hospital in that year), **emslevel** (each hospital's EMS level; Standby, Basic, or Comprehensive), **stations** (how many patient-accessible ED stations were present for the hospital in that year), **diverthours** (the number of diversion status hours for the hospital in that year), and **admitrate** (the admission rate for the hospital in that year, calculated as **admit** divided by **totalvisits**). To gain insight into the ambulance diversion phenomenon, the present study aimed to answer this research question: What factors are related to the number of California hospital ambulance diversion hours between 2013 and 2015?

Methods

Data Cleaning and Processing

To answer the research question of interest, the dataset was first cleaned and processed. Namely, due to the small number of hospitals associated with the Level I, Level II, and Level III trauma center designations, the trauma center designation variable was converted to a factor with two levels (0/Level III, the reference level, and Level II/Level I). This grouping was chosen based on Melnick (2025), which describes Level I as the maximum trauma care level. Similarly, due

to the low number of hospitals with a Standby or Comprehensive EMS level, the EMS level variable was converted to a factor with two levels (Standby/Basic, the reference level, and Comprehensive).

Next, since 14 of the 25 counties in the dataset each contained just three or fewer hospitals, the counties were consolidated into their corresponding 10 regions according to the categorization scheme from State of California (n.d.). In doing so, a new factor variable **region** was created with levels Superior California (the reference level), North Coast, San Francisco Bay Area, Northern San Joaquin Valley, Central Coast, Southern San Joaquin Valley, Inland Empire, Los Angeles County, Orange County, and San Diego - Imperial. In addition, a variable **time** was added to the dataset representing the measurement occasion number for each hospital. Furthermore, the variable **year** was treated as continuous rather than categorical due to approximate visual linearity in the mean trajectory for the ambulance diversion hours from 2013 to 2015.

References

- Melnick, Glenn. 2025. "Hospital-Based Emergency and Trauma Care—the Expanding Epicenter of the US Healthcare Delivery System." *Healthcare* 13 (12): 1424. <https://doi.org/10.3390/healthcare13121424>.
- State of California. n.d. "Regions." Accessed March 23, 2026. <https://census.ca.gov/regions/>.